

KUBERNETES

Inhoud

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1 Introduction

This guide is intended to help you deploy a Kubernetes cluster through the Previder Portal. It provides a walkthrough of the deployment process and explains the available configuration options.

Throughout this guide, we will highlight the choices available during cluster creation and provide guidance where applicable, enabling you to configure a Kubernetes cluster that best fits your requirements.

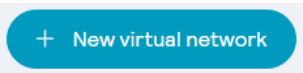
2 Preparation

Note the IP address from which you will connect to the Kubernetes platform. You can find it via:
<https://ip.previder.nl>

3 Deploy a Virtual Network

Before creating the Kubernetes cluster, first create a Virtual Network. This network is required during the Kubernetes cluster deployment process.

In the Previder Portal, go to the “IaaS” menu, then select “Virtual Networks” and click “New Virtual Network”.



Enter a name in the “Name of the network” field and select the type of network you want to use. In this guide, we choose an Isolated VLAN; however, if you plan to use other Previder services later such as Managed Firewall, StaaS, or Colocation then select a Cloud VLAN instead.

The network name used in this guide is: “Kubernetes Demo”.

Configure your network

Name of the network *

Kubernetes Demo

Customer reference

Customer reference

Group

Select group

Tags

Add item

Shared with

This network is not shared with another customer

Type of the network *

Isolated Network

Isolated Network
A free of charge network to use in the entire Previder IaaS environment.

Cloud VLAN
A network with a fixed monthly fee which can be used to link to other services of Previder, like Managed Firewall, STaaS or Co-location.

VLAN Trunk
A free of charge network with one, or multiple non-public Cloud VLANs. This network is configured as a VLAN trunk.

Shared networks

It is possible to share a network with a super or sub customer. The super or sub customer can use the network, but cannot make any changes to it.

Share network

Wait until the network has the status “READY”.

Name ↓↑	Group	Type ↓↑	Status
Kubernetes Demo		Isolated Network	READY

3.1 Choosing between DHCP pool and an IP Pool.

When deploying the Kubernetes cluster, you can choose between a **DHCP Pool** and an **IP Pool**.

- **DHCP Pool:** IP addresses are assigned automatically by a DHCP server. This offers flexibility and reduces administration. However, IP addresses may change after redeployment, lease expiration, or network changes, which can cause connectivity or cluster stability issues.
- **IP Pool:** IP addresses are assigned from a predefined range. This provides a predictable network configuration and ensures nodes retain their assigned static IP addresses after redeployment, simplifying firewall, DNS, and monitoring management.

Recommendation: For production environments, we recommend using an IP Pool for better stability and predictability. A DHCP Pool is more suitable for test or development environments.

it starts here.

3.2 Create IP pool

If you do not want to use an IP Pool and prefer to use a DHCP Pool instead, you can skip this step.

In the Previder Portal, go to the “Virtual Networks” menu, navigate to the network you just created, in our case “Kubernetes Demo”, and under the “IP Addresses” section, click “New Address Pool”.

×

Create address pool

Type

☒ IPV4 ☐ IPV6

Start address	End address	Network mask
192.168.1.50	192.168.1.150	255.255.255.0
Gateway address	Nameservers	
192.168.1.1	80.65.96.50	62.165.127.222

* Please note that Previder does not provide DHCP services in your network.

Cancel

+ Save

After configuring the Address Pool, click "Save" to apply your changes before proceeding to the next step.

4 Deploy the Virtual Firewall

In the Previder Portal, go to the “IaaS” menu, then select “Virtual Firewalls” and click “New Virtual Firewall”.

Under “Network”, select the Virtual Network you just created, in our case “Kubernetes Demo”. Under “LAN Addresses”, enter the IPv4 CIDR you want to use. Also configure the preferred DHCP range start and end addresses. If you want to use an IP Pool, disable DHCP. The virtual firewall will then automatically assign an available IP address from the IP Pool of the selected Virtual Network.

1. Create a virtual firewall

✓ Previder



- A virtual firewall developed by Previder

2. Network settings

Network * Kubernetes 	LAN Address ⓘ * 192.168.1.1/24	DHCP ☒ Enable DHCP server in the network	DNS ☒ Enable DNS in the network
DHCP range start address ⓘ * 192.168.1.50	DHCP range end address ⓘ * 192.168.1.150	Local domain name ⓘ * int	Nameservers ⓘ * Add item + 80.65.96.50 × 62.165.127.222 ×
Allow ICMP for WAN ☒ Allow ICMP to your WAN interface	Allow ICMP for LAN ☒ Allow ICMP to your LAN interface		

3. General settings

Firewall name * Kubernetes Demc	Group Select group	Termination protection ☐ Protect your firewall from accidental deletion
--	---	---

Cancel

+ Create

If you have filled in all the required fields, click Create and wait until the state changes to READY.

	Kubernetes Demo	Kubernetes Demo	192.168.1.1/24	62.165.77.57, 2001:1690:100:3f00:0:0:1:17c	Previder	READY
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5 Deploy Kubernetes cluster

Now it is time to deploy the Kubernetes cluster. In the Previder Portal, go to the “Kubernetes” menu, select “Clusters”, and click “New Cluster”.

5.1 Choose cluster type

Step 1, select the cluster type and choose whether the nodes should be deployed in a single datacenter or distributed across multiple datacenters. For this guide, we will use the “Express (PDC1 & PDC2)” cluster type.

1. Choose type

☒ Express

- Web or application hosting
- Selection for 1 datacentre
- Limit to 5000 IOPS

PDC1 & PDC2

Performance

- Database hosting
- Selection for 1 datacentre
- Limit to 20000 IOPS

Make a choice

Premium

- Company-critical hosting
- 2 Data centers
- Limit to 20000 IOPS
- Real-time replication

Make a choice

it starts here.

5.2 Setup your control plane

Step 2 is to choose whether the control plane should be high available (HA). For this guide, we will not enable "High availability" or the "Advanced Setup" option. If you choose to enable HA, three control plane nodes will be deployed across one or multiple datacenters, depending on the selected cluster type.

By enabling the "Advanced Setup" option, you can customize the resources allocated to the control plane nodes. By default, each control plane node is provisioned with 2 vCPUs, 4 GB of memory, and 25 GB of storage.

2. Setup your control plane

High availability ⓘ
☐ Disabled

Advanced setup
☒ Enabled

CPU cores
- 2 CPU +

Memory
- 4 GB +

Storage
- 25 GB +

5.3 Setup your worker nodes

Step 3 is to choose how many worker nodes should be deployed. You can also enable autoscaling and, similar to Step 2, use the "Advanced Setup" option to define the resources for each worker node, or select one of the available presets: Small, Medium, Large, or Extra Large.

When autoscaling is disabled, specify how many worker nodes should be installed. When autoscaling is enabled, specify the minimum and maximum number of worker nodes. More information about autoscaling can be found in the help under the "Autoscaling" chapter: [Autoscaling documentation](#).

3. Setup your worker nodes

Number of worker nodes
- 1 +

Enable autoscaling ⓘ
☐ Disabled

Choose a default size or create an advanced setup.

Advanced setup
☐ Disabled

☒ **Small**

- CPU cores: 2
- Memory: 4GB
- Storage: 25GB

Medium

- CPU cores: 4
- Memory: 8GB
- Storage: 50GB

Large

- CPU cores: 4
- Memory: 16GB
- Storage: 100GB

Extra large

- CPU cores: 8
- Memory: 32GB
- Storage: 250GB

5.4 Global setup

Step 4 is to choose a cluster name and optionally enable termination protection, which makes it harder to accidentally delete a cluster. You can also enable “Automatic updates”.

4. Global setup

Cluster name *

kubernetes-demo

Group

Select group

Termination protection ⓘ

☐ Disabled

Automatic updates ⓘ

☒ Enabled

> Advanced options

If you choose the “Automatic updates” option, Previder will ensure that your cluster is periodically updated to the latest stable version of Kubernetes.

If you prefer to manage the version yourself, you can disable "Automatic updates" and select the Kubernetes version you want to install and decide when to upgrade to a newer version.

The version can only be upgraded; downgrading a Kubernetes cluster is not possible.

Automatic updates ⓘ

☐ Disabled

Kubernetes version ⓘ *

Make a choice

Under “Advanced options”, you can configure an authentication setup, allowing you to log in via an external identity provider such as Keycloak.

Advanced options

Authentication config

5.5 One-time setup

Step 5 is to choose the Container Network Interface (CNI) that will be configured for the cluster. For this guide, we will use “Cilium”. You can also choose not to pre-install a CNI and install and manage your preferred CNI yourself.

Also select the network in which your Kubernetes cluster should be available. This network must meet certain requirements before a cluster can be successfully deployed. If all requirements are met, ensure that the following items are checked: "DHCP or IP Pool configured with DNS" and "DNS resolution available".

5. One-time setup

Choose a CNI ⓘ	Network requirements ⓘ	Selected network
Cilium	☒ DHCP or IP Pool configured with DNS' ☒ DNS resolution available ⓘ The selected network must have DHCP enabled or a static IP pool in which DNS is available with internet connectivity.	Kubernetes Demo

Next, configure the Virtual IP (VIP) address on which the cluster should be reachable. This address must belong to the selected network but must not be part of a DHCP pool. For this guide, we will use the IP address 192.168.1.10.

You can also configure additional endpoints (FQDNs) through which the cluster can be reached. These endpoints will be added as Subject Alternative Names (SANs) to the cluster certificates alongside the VIP. An example endpoint would be cluster.customer.com. For this guide, we will use the public WAN IP address of the Virtual Firewall.

VIP ⓘ *

Add item +

192.168.1.10 ×

Endpoints ⓘ

Add item +

62.165.77.119 ×

After reviewing the configuration, click the “Create” button to deploy the Kubernetes cluster.

it starts here.

6 Connect to the Kubernetes Cluster

In the following steps, we assume that `kubectl` is already installed on your system. If it is not installed, you can refer to the official Kubernetes documentation for installation instructions.

<https://kubernetes.io/docs/tasks/tools/>

6.1 Create a NAT Rule to Expose the Kubernetes Cluster

Go to “Virtual Firewalls” and select the firewall you created earlier. Next, navigate to “NAT Rules” and click “New NAT Rule”.

Create a NAT rule for Kubernetes Demo

Description *	Source CIDR ⓘ
Kubernetes	0.0.0.0
☒ Active rule	
WAN port *	Protocol *
6443	TCP
Destination IP *	Destination port *
192.168.1.10	6443

Fill in a description, in this guide, we will use “Kubernetes”. As WAN port and destination port, we will use port 6443, the port of the Kubernetes API server.

We leave the Source CIDR empty in this demo environment. In production, you should restrict which source IP addresses are allowed to access the cluster. Never expose port 6443 worldwide.

Also fill in the Destination IP; this is the VIP address selected during the cluster deployment. Once all required fields are completed, click “Create”

Kubernetes Demo ×

⋮ Properties ⚙️ NAT Rules

+ New NAT rule

Description ↓↑	Source	Port	NAT Destination	NAT Port	Active ↓↑	
Kubernetes		6443	192.168.1.10	6443 / TCP	Yes	✎ 📄 🗑️

it starts here.

6.2 Download the Kubeconfig File

You can find the kubeconfig you need in the Previder Portal under the “Kubernetes” menu. Select “Clusters”, choose the cluster you created, and navigate to the “Endpoints” tab. Click the download button for the kubeconfig you want to download. For this guide, we will use the kubeconfig with IP address 62.165.77.119.

kubernetes-demo					×
☰ Properties	🖨 Nodes	📄 Endpoints	📄 Certificates	☰ History	
Endpoint / VIP					
192.168.1.10					📄
62.165.77.119					📄

6.3 Access the Cluster Using kubectl

You should place the kubeconfig file in the default location used by kubectl. This normally is the .kube directory in your user home folder. If the directory does not exist yet, create it and move the file there, naming it config. For this guide, the kubeconfig should be placed at ~/.kube/config so that it is automatically detected by kubectl without requiring additional configuration.

When everything has been set up correctly and you run kubectl get nodes, you should see the following output:

NAME	STATUS	ROLES	AGE	VERSION
prev01-kubernetesdemo-001705-jmjwh	Ready	control-plane	2d21h	v1.35.4
prev01-kubernetesdemo-001705-nn976-2c4tl	Ready	<none>	2d21h	v1.35.4
prev01-kubernetesdemo-001705-nn976-vqrb1	Ready	<none>	2d21h	v1.35.4

On our GitHub page under **kubernetes-examples**, you can find several useful guides. These include, for example, how to configure an Internal Load Balancer setup with L2Advertisement, as well as how to connect STaaS using the STaaS with NFS Subdir provisioner.